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Subject dependent speech verification approach for assistive special education

Umut Zeki[1] · **Tolgay Karanfiller**[2] · **Kamil Yurtkan**[1,3]

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Abstract

The developmental, characteristics and educational competencies of students who need special education are developing slowly in compared to their agemates. This is because their expressive language is different. In order to overcome these challenges, assistive technologies can be used under the supervision of the teachers. In this paper, a person dependent speech verification approach is proposed for special education students. The system verifies the speech of special education students in various ways. Convolutional Neural Network (CNN) is employed for the classification task. Audio signals that are collected as datasets are accomplished by collecting samples from the real education centers involving special education students. For each subject, a different CNN is trained. Obtained audio signals undergo a frequency domain transform, and then their spectrograms are computed. The spectrogram images of every audio sample are then processed as inputs to the CNN. In this way, better representations of the audio signals are achieved where the spectrogram images of the audio files of different subjects are discriminable. This is also the result of special education students' personal and unique speaking styles. The proposed approach is tested on the dataset that is constructed by real subject recordings. The system achieves promising results by performing comparable recognition accuracies of around 96%.

Keywords Speech verification · Special education · Deep learning · Convolutional neural Networks · Speech recognition

✉ Umut Zeki
umut_zeki@hotmail.com

[1] Software Engineering Department, Rauf Denktaş University, Northern Cyprus Via Mersin 10, Girne St. Surlarici, Nicosia, Turkey

[2] Faculty of Education, Cyprus International University, Northern Cyprus Via Mersin 10, Haspolat, Nicosia, Turkey

[3] Artificial Intelligence Application and Research Center, Cyprus International University, Northern Cyprus Via Mersin 10, Haspolat, Nicosia, Turkey

1 Introduction

Speech recognition and verification are frequently confused. On one side, "speech recognition" is the skill of being able to identify spoken words. On the other side, speech verification uses speech recognition to verify the accuracy of the spoken language. Speech recognition is a simple task for an adult human, whereas it is a challenge for the modern computer. The challenge is that everyone's pronunciation varies. Speech verification is the solution to this problem; that uses the speech and verifies if the desired word is spoken. Speech recognition technology is used in many applications today. Such technologies can also be found in educational applications. These days, there is a growing trend and demand for educational software created for people with special needs. However, successfully implementing the aforementioned technologies in developed educational applications is a difficult procedure. Because individuals who have special needs have challenges using the previously mentioned applications via speaking (İlker & Melekoğlu, 2017). This makes speech recognition task difficult. However, in order to get satisfied results while studying with people who require special education, it is preferable to work with data collected directly from them, instead of working with existing datasets. Because they may have problems with making sounds or problems of colloquial stuttering, expressed in speech flow disturbances, partial word repetitions, or prolonged sounds and syllables. For all these reasons, speech recognition systems designed for individuals with normal development are not successful enough in special education.

Based on above mentioned problems, we propose a novel person-dependent speech verification approach designed for students who has special needs. Our system gets the user's speech samples once using them for the first time as a dataset. Then, the verification step is done. Convolutional Neural Network (CNN) is used as classifier. On the other hand, the pre-processing advantages are inherited (Saha, 2018). The CNN employed is created and trained from the scratch. The dataset used is also an originality of the paper. Audio signals that are collected as datasets are accomplished by collecting from the special education students by recording the audio samples in which they pronounce digits. For each subject, a different CNN is trained. Another level is to change the audio representation. Here, the audio signal undergoes a frequency domain transform, and then its spectrogram is computed. The spectrogram images of every audio sample are then processed as inputs to the CNN. With this way, a better representation of the audio signal is achieved where the spectrogram images about the audio files of different subjects are discriminable. The proposed system is unique because there is no study encountered in the literature for special education which does person-dependent speech verification. Therefore, these technologies can be employed in other special education applications to accurately check the student's speech. As a result, the students with special needs will benefit from similar assistive technologies. The models are simulated and the performance evaluations are done accordingly.

The development of assistive technologies is utmost important in several aspects. For example, enhancing the teacher's performance with an assistive

technology is important. On the other hand, by using assistive technologies, a digital archive can be formed where the progress of the students can be tracked. Another importance is that the students themselves can use automated applications under the supervision. Regarding all these aspects, the proposed approach is solving the automatic audio verification tasks where it can be adapted to any related assistive technology to be used in special education domain.

Considering the marginality of the speech of individuals with special needs, the proposed approach will fill in this area. The proposed approach has been developed for a possible mobile application's speech verification module. In this way, when the user uses the speech recognition feature, the evaluations will be done using the subject's own dataset, and the speaking problems can be easily overcome. The main contribution of the paper is that it is one of the proposed approaches applied on real data that come from special education students and that will be a positive addition to the related literature. Also, its person-dependency using the spectrogram levels before the CNN computations are applied to these original data. The approach achieves promising results by performing comparable recognition accuracies of around 96%.

2 Related works

In applications using machine learning and computer vision, speech recognition presents a significant problem. Numerous machine learning techniques have been employed to address the issue of speech recognition in the literature.

The first speech recognition mechanism was developed in 1920. It was a celluloid dog who came out from his base when he heard and recognized his name (David & Selfridge, 1962). During the 1950s, Audrey used speech recognition technology to recognize digits (Biddulph & R & Balashek S, 1952). A tool called the Shoebox was presented by IBM at the 1962 Seattle World's Fair. Besides being able to perform mathematical functions, it could also recognize the digits and 16 spoken words (Shaikh, 2023). Significant improvements are reported later then in the field.

Deep learning, on the other hand, has improved voice recognition to the point that it can be used outside of highly regulated contexts (Geitgey, 2016). The suggested Intelligent Phishing Detection System (IPDS) by Adebowale et al. had a classification accuracy of 93.28%. The technique can screen harmful websites based on behavioral characteristics obtained from earlier data sets (Adebowale et al., 2020). Bingol and Aydogmus proposed a DNN Pruning Deep Neural Network (p-DNN), which was compared to different classification techniques, in 2020. A robot manipulator was created and used to implement the HRI framework. The robot begins to listen to the dialogue when it hears the term "KUKA" (Vickers, 2017). Moreover, in 2019, Huang et al.'s approach to identifying speech emotion in spontaneous speech takes into account both verbal and nonverbal speech sounds. Experimental results showed that the proposed approach procured a 52.00% detection accuracy that exceeded classical methods (Huang et al., 2019).

In contrast to conventional classification techniques, CNN, a sub-branch of Deep Learning, can take an input image and separate its many aspects and objects

with less pre-processing. In this field, there have been a lot of studies done on speech recognition. For example, Jing et al. (2019) suggested two voice recognition algorithms based on CNN to improve speech accuracy. A dynamic adaptive pooling (DA-pooling) algorithm is suggested for the CNN model's pooling layer. In a full-connected layer, a dropout strategy based on sparseness is presented to overcome standard dropout hiding neuron nodes randomly. The results suggest that adding a DA-pooling algorithm and improving the dropout sparseness technique to CNN-based speech recognition will boost the sparseness of speech recognition. To classify emotions in speech, Zhao et al. (2019) presented 1D and 2D CNN LSTM networks. Raw audio recordings and log-Mel spectrograms were used to investigate how local similarities and global contextual knowledge may be gleaned from them. On the Berlin EmoDB of speaker-dependent and speaker-independent experiments, the 2D CNN LSTM network achieves 95.33% and 95.89% recognition accuracies, respectively, which is far better than the conventional approaches. Solanki and Pandey (2022) proposed a deep convolution neural network architecture for identifying polyphonic music instruments. The network is designed to calculate an arbitrary number of instruments for fixed-length music. After 60 epochs, the study finds an exceptional result of 92.80% precision. In 2018, Nagajyothi and Siddaiah (2018) presented an airport inquiry framework based on ASR. The database is created based on the most frequently asked questions in an airport inquiry. Because of its high efficiency, CNN has been used for database training and analysis. There are several applied methods for speech recognition. But there are very few studies that use spectrograms for speech recognition. Musaev et al. (n.d.) has studied speech recognition by using spectrogram images of Uzbek spoken digits. Spectrogram images from speech signals were generated and they were used for deep CNN training. As a subset of speech recognition, there are no study has been encountered related to speech verification using CNN in the literature. However, numerous research in various disciplines have used CNN to verify the correctness of a picture, a face, a fingerprint, or a traffic sign. For instance, a study proposed by Mohapatra (Hafemann et al., 2017) was conducted in the realm of biometrics in 2019 and used CNN to verify offline handwritten signatures. On the other hand, a unique end-to-end method for text-dependent speaker verification is described in the work on speaker verification that Shi-Xiong Zhang published in 2016 (Zhang et al., 2016). Additionally, there are several studies in the area of face verification. The approach for unbounded face verification presented by Jun-Cheng Chen that year uses convolutional features and is evaluated on both the conventional Labeled Face in the Wild (LFW) dataset and the freshly released IARPA Janus Benchmark A (IJB-A) dataset (Chen et al., 2015). There are many speech recognition studies on individuals in need of special education. In 2011 Garret J. T. employed speech recognition software as an alternate treatment to help people with physical limitations write more fluently (Garrett et al., 2011). The findings also showed that when utilizing speech recognition rather than word processing, people with physical limitations wrote longer drafts. Ok M. W. investigated how students with high incidence impairments used speech recognition (SR) technology in grades 4–8 and how students and teachers perceived utilizing SR as a writing process in 2020 (Ok et al.,

2022). Overall, research on speech recognition applications still faces challenges. To achieve robust recognition systems, significant research studies are ongoing.

The implemented approach in this study is a person-dependent speech verification system using CNN. Rather than recognition, the proposed approach performs verification task. The following is a brief explanation of the differences between these two methods:

Speech verification answers: Is this digit x? Whereas, speech recognition answers: What is this digit?

The proposed approach utilizes the information in the input audio signal by transforming it into the frequency domain. This is achieved by spectrogram transformation of the audio signal. The transformation is summarized in Sect. 3.2.

3 Speech verification for special education

The speech recognition system is capable to translate the spoken words into text. Text can be either in terms of words or sequences of words or it could be in syllables form. Speech recognition is one of the challenging tasks in machine learning. We know that human speech signals are very varied due to differences in speaker characteristics and speaking styles, and that they can even incorporate environmental noises (Nidhi & Satish, 2019). Speech verification is a sub-discipline of speech recognition. The system attempts to validate a spoken digit, as the name implies. It is a direct comparison. The system's goal here is to validate whether the spoken digit is correct rather than to correctly recognize it. Figure 1 depicts this scenario.

A spectrogram is a visual representation of a signal's frequency spectrum as it changes over time. A graph with two geometric dimensions is a typical format: one axis indicates time, while the other axis shows frequency. In this study, spectrograms are constructed from a time-domain signal by computing the Fourier Transform (FT) from the time signal. An FT is a mathematical transform that decomposes functions based on space or time into functions by using spatial or temporal frequency. FT is mainly used to create a spectrogram. It's a well studied transformation in recent studies. Considering the signal in the time domain, it is sampled and decomposed into windows that may overlap. Then, the frequency analysis are done for every window based on FT. Each chunk is then represented as a vertical line in the image, which represents a magnitude versus frequency measurement at a certain point in time. These spectrums or time plots are then "placed side by side" to take a picture or a three-dimensional surface, or slightly overlapped in various



Fig. 1 A generic digit verification system

ways, which is a process known as windowing. The Short-Time Fourier Transform (STFT) squared magnitude of the signal is determined in this technique (Spectrogram n.d.). STFT (Sejdić et al., 2009) is a Fourier-related transform that is used to determine the sinusoidal frequency and phase content of small sections of a signal as they vary over time. In practice, STFTs are computed by dividing a larger temporal signal into equal-length segments and computing the FT independently on each shorter segment. The Fourier spectrum is presented on each shorter section. The changing spectra as a function of time are then displayed using a spectrogram or waterfall plot, which is commonly used in Software Defined Radio (SDR)-based spectrum displays. Full bandwidth displays covering the entire spectrum of an SDR often use Fast Fourier Transforms (FFTs) with 224 points. Simply put, the function to be transformed is multiplied by a window function that is nonzero for just a short period in the continuous-time situation. As the window is moved down the time axis, the FT (a one-dimensional function) of the resulting signal is taken, resulting in a two-dimensional representation of the signal. Equation 1 shows the transformation.

$$\mathbf{STFT}\{x(t)\}(\tau,\omega) \equiv X(\tau,\omega) = \int_{-\infty}^{\infty} x(t)w(t-\tau)e^{-iwt}dt \tag{1}$$

The window function is w(T), which is usually a Hann window or a Gaussian window that is centered to zero. The signal to be transformed is x(t). W is the frequency, and w is the window function. x(T,W) is the Fourier transform of x(t)w(t-T), which is a complex function that represents the signal's phase and magnitude with time and frequency (Allen, 1977). For example, a one-dimensional waveform image is turned into a spectrogram image as in Fig. 2, allowing it to be easily recognized from other sounds. In the proposed approach, every spoken digit was captured, and then the audio signals were transformed into spectrograms using the STFT method.

CNN is designed basically for digital images and adapted to take matrices as inputs for the input layers. Using spectrogram images of the audio input supports this input layer flexibility of the network as well as enhancing the signal representation. Otherwise, the first layers must be modified in order to support one-dimensional audio signals as inputs. Furthermore, using a spectrogram as the input of CNN reduces it to an image classification problem. This image represents the

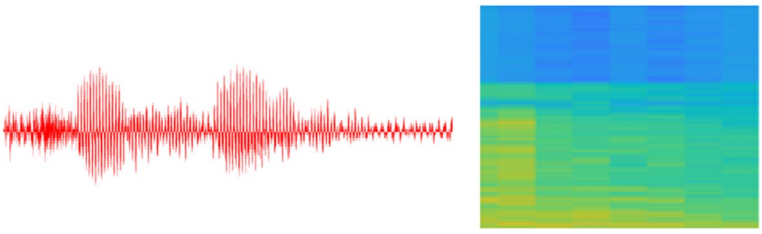


Fig. 2 Generated waveform image from an audio file of the student's pronunciation of zero is converted to a spectrogram image

spoken digit from left to right promptly. It can also be used to verify a phrase that is written from left to right, by identifying the hidden characters (Katrik, 2020).

CNN is a deep learning class that requires substantially less pre-processing and produces better outcomes as the number of training sessions grows (Gilu & Bhaskaran, 2020). Figure 3 depicts the network architecture of the voice recognition CNN technique. The processing in CNN starts with feature extraction layers and ends with fully linked Neural Network classification layers. Using many layers enhances recognition accuracy while being insensitive to tiny geometric alterations of the input images. CNN is successfully utilized on real-world data for classification jobs due to its robustness.

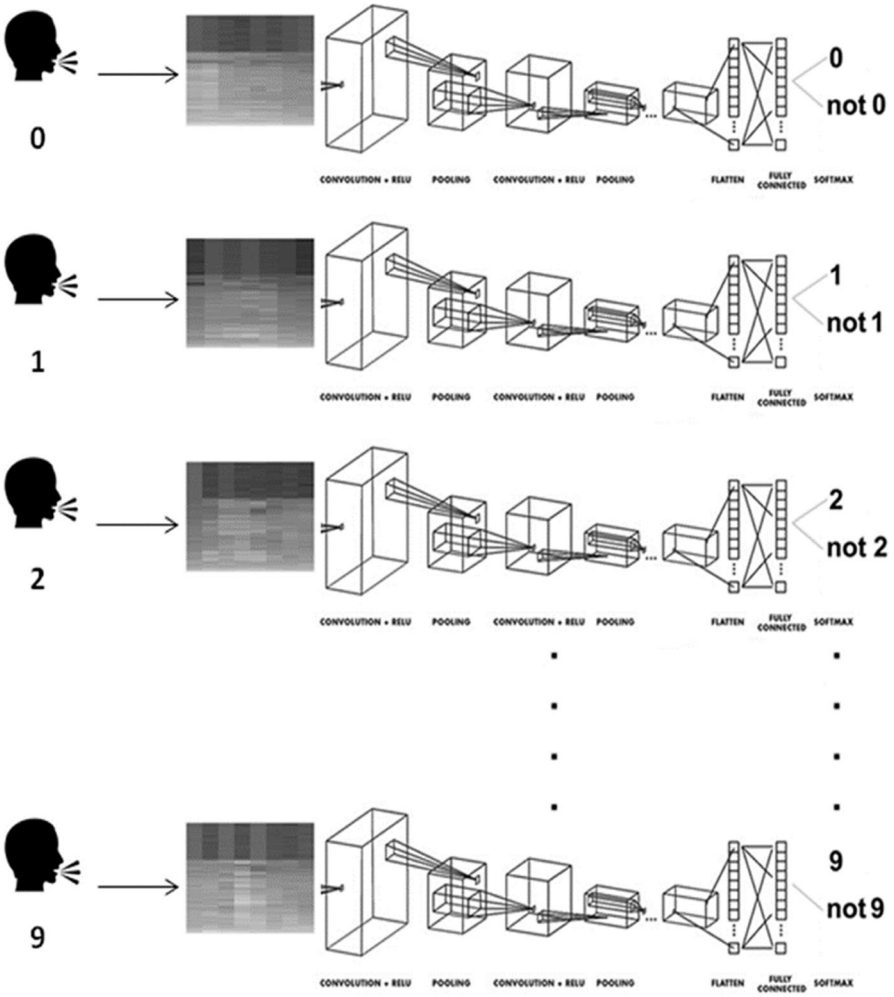


Fig. 3 The architecture of our proposed approach: Ten different person-dependent CNNs

4 Proposed approach

The proposed method is being developed for a special education application to teach numerals. It seeks to recognize the digit pronounced by a user in a short period of time with high precision. In one area of the application, the user should respond to the query by saying. As a result, before beginning to learn a digit, each student's sound samples should be collected as inputs. For example, if digit 2 is to be taught, the application instructs the pupil to speak it numerous times. These data are subsequently employed in the training of CNN, together with the student's sound, while learning the given digit in the application (Fig. 3). What the student says at the beginning of the application is used as a training sample, and what the student says throughout the application is used as a test sample. In this method, person-dependent training is carried out. This means that each subject must have its own CNN. There should be $n \times 10$ CNN structures for every n subjects. This is the outcome of having ten different spectrograms to verify for each digit (0–9). Figure 3 depicts the general technique, where 10 (0–9) CNN are constructed for a single subject.

4.1 Dataset

A special education expert visited Algm Special Education Center to create the datasets. The speech samples were collected by ten mildly impaired special education students. To avoid distractions while collecting students' speech samples, a quiet and plain room was set up. To avoid data loss, a dedicated voice recording device was employed to gather and transfer information to the computer. Also a special education expert was present during the data collection process. The students voice is recorded during their spellings of the digits from 0 to 9 (0–9). They have entered to the room one by one to avoid any interference. The process was carried out ten times for each student. A break was given as necessary because some of the pupils became easily distracted and tired of repeating the same figures. Since some students were excited when they first entered the class, or because they had difficulties repeating the numbers accurately, they were requested to repeat the numbers from some sets. Overall, one thousand sampler acquired from the total of ten students with slight mental retardation. Audio signals were transformed to RGB spectrogram images after gathering sample forms. As a result, image resolutions were reduced during designing by considering the inverse ratio between proper estimation rate and rapid running. Then the images undergo grayscale transformation. Figure 4 shows some examples. The dataset is then divided into two parts where the first part includes 3600 samples (10 users $\times$ 10 sets $\times$ 9 false digits $\times$ 4 samples) for training, and the part two includes 400 samples (10 users $\times$ 10 sets $\times$ 1 true digit $\times$ 4 samples) for testing. For convenient use, all the images are stored in an appropriate file format.

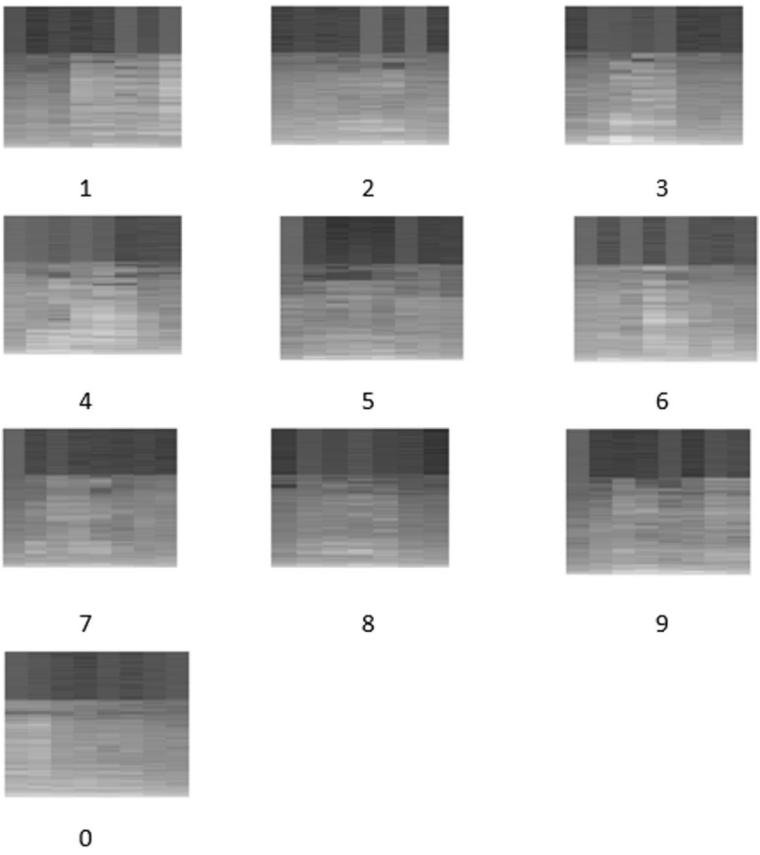


Fig. 4 Speech samples of digits (one set) of one subject are converted to spectrogram images

4.2 System architecture

The transformed signals undergo preprocessing steps. To use the spectrograms properly, the preprocessing steps are compulsory.

Firstly, because of the audio files of each student containing all the digits in one file, they are cropped to separate them into single digits without any space at the beginning and the end. Then, each audio file is transformed into a spectrogram. Because of having spectrogram files in RGB format with high resolution; they are then converted into grayscale(black and white) images with lower resolution. This process is done to spped up the system.

The data of each student was put in a separate folder. Ten subfolders have been added for each digit and named as 0,1,2…9 respectively. In each subfolder, two more folders were created and named as true and false. Spectrogram samples of the digits that the system needs to verify are placed in the true folder. The rest of the digits are put in the false folder. As it is illustrated in Fig. 5, if thestudent "Furkan"'s digit 0 is to be verified, sound sample spectrograms of 0 are placed in



Fig. 5 The illustration of file and folder structure is used in the proposed algorithm

the true folder and sound sample spectrograms other than 0 (1–9) are placed in the false folder.

A CNN is made up of numerous layers, each of which transforms a volume of activations across a differentiable function. While building the network, we used a total of 15 layers. These are Convolution Layer, Pooling Layer, Relu Layer, Softmax

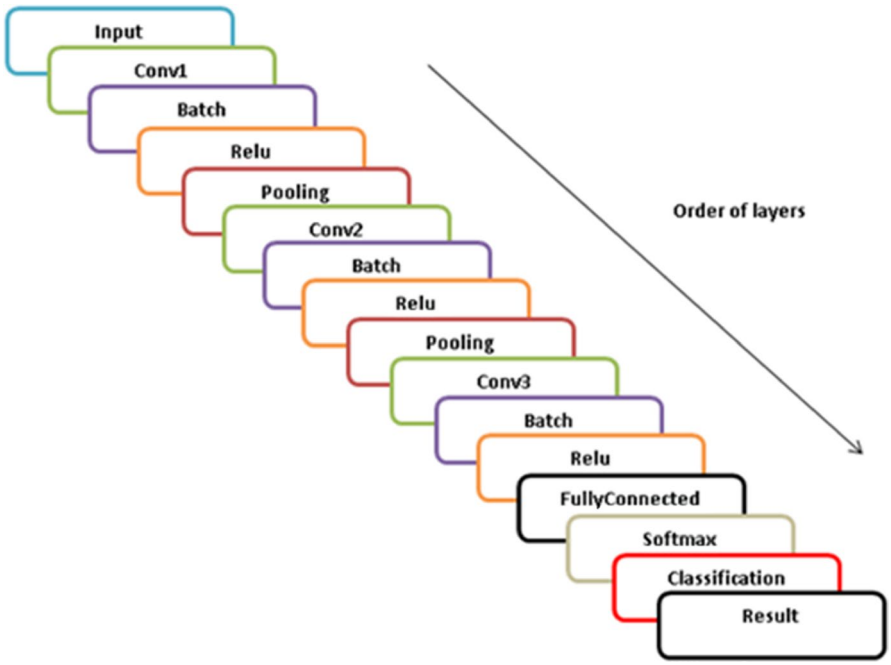


Fig. 6 Illustration of the Convolution Neural Network model used

Layer, Classification Layer, Image Input Layer, and Batch Normalization Layer. The architecture of the system is illustrated in Fig. 6.

CNN's first layers function as feature extractors, and the first block of that type of neural network builds the feature. It applies template matching by utilizing convolution filtering processes. The first layer filters the image using convolution kernels and returns "feature maps" that are normalized using an activation function and/or shrunk. The audio signal undergoes a frequency domain transform and then its spectrogram is computed. The spectrogram images of every audio sample are then processed as inputs to the CNN. So, the spectrogram is used in the input layer like input 2-D images to a network. Its dimension is 100×186x1 with 'zero center' normalization. The resolution of the supplied image is 100×186, and 1 denotes a grayscale image. The input images are then sent through a series of convolutional filters, each of which activates different features in the images. The number of filters in this case is eight, and the filter size is 3×3. Convolutions are also used with stride [1 1] and padding [0 0 0 0]. A Batch Normalization Layer is used after convolution to equalize each input channel across a mini-batch. This layer is utilized between the Convolutional and Relu Layers to accelerate CNN training and reduce susceptibility to network initialization. The fourth layer, Relu Layer, serves as the activation layer. A Relu Layer applies a threshold operation to each element of the input, setting any value less than zero to zero. Following that, we must do downsampling by dividing the input into rectangular pooling regions and determining the maximum of each zone by adding a Max-Pooling Layer with pool size [2 2] and stride [2 2]. In the second block, all layers except the input layer are repeated. The Convolution Layer is the only difference here. Its specifications differ from the first. This time, a 2-D Convolutional Layer is built with 32 filters of size [3 3] and'same' padding. During training, the software finds and puts the size of the zero paddings, so that the output layer has the same size as the input. In the last block of Convolution Layer like the second block, Batch Normalization Layer and Relu Layer are repeated. Later, the fully connected layer after the Relu Layer performs the classification. Then, multiplies the input by a weight matrix and to a bias vector with an output size of 2. The last two layers are output layers. The first one is the Softmax Layer and the last one

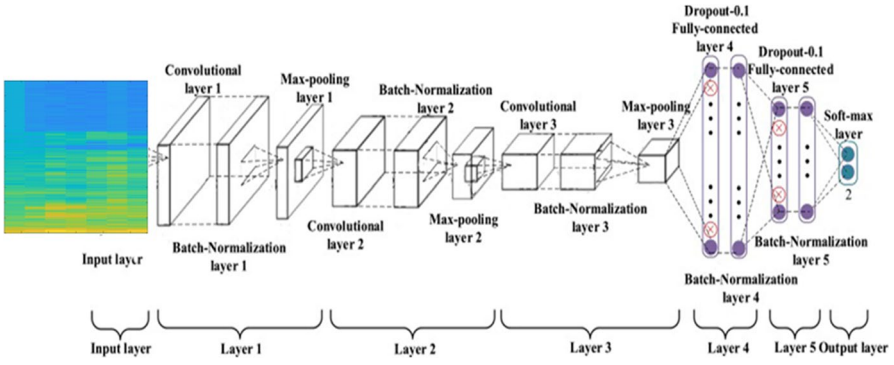


Fig. 7 The architecture CNN approach for one subject and one digit

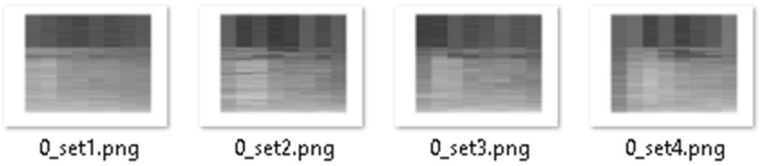


Fig. 8 True(Test) class of a user for 0

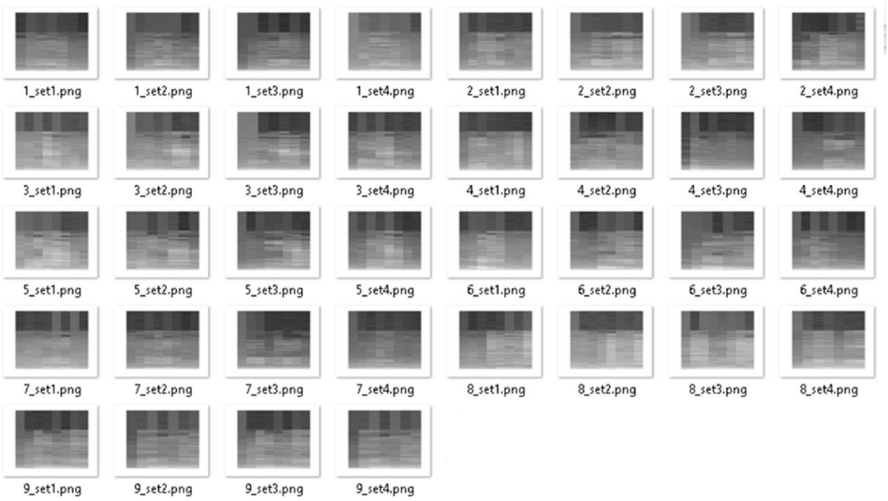


Fig. 9 False(Training) class of a user for 0

is the Classification Layer. Figure 7 shows the architecture of the CNN approach used in details for one subject and one sound signal of one digit (Bishop & Nasrabadi, 2006; Ioffe & Szegedy, 2015; Krizhevsky et al., 2017; LeCun et al., 1989, 1998; Murphy, 2012; Nagi et al., 2011; Nair & Hinton, 2010; Shijie et al., 2017).

5 Results and discussions

5.1 Experimental setup

A deep learning library is used that is a network architecture to implement CNN architecture in the simulator. The pixel sizes of the images that are trained are small enough to maintain acceptable processing time whereas large enough not to reduce the recognition quality. There are 4000 samples which are 3600 from training and 400 from testing by a training dataset separately.

The samples that the system would train, were put in the True class (Fig. 8); and the samples that the system would test, were put in the false class (Fig. 9). For example, if the target is to verify whether the pronounced digit is 5 or not, the true class

Table 1 Training, validation, and learning rate information

Name	Value
Training time	31 s
Number of Epochs	10
Validation Frequency	30 iterations
Learning rate	0.001

Table 2 Training accuracy of each subject/digit

Subject/Digit	0	1	2	3	4	5	6	7	8	9
subject1	90%	100%	100%	90%	90%	100%	100%	100%	100%	100%
Subject2	100%	90%	100%	100%	100%	90%	100%	90%	90%	100%
Subject3	100%	100%	90%	100%	100%	90%	100%	90%	100%	100%
Subject4	100%	100%	100%	90%	90%	100%	100%	100%	90%	100%
Subject5	100%	100%	100%	100%	100%	90%	100%	100%	100%	90%
Subject6	90%	100%	100%	90%	90%	100%	100%	100%	100%	100%
Subject7	90%	90%	100%	100%	100%	100%	90%	90%	100%	100%
Subject8	100%	100%	100%	90%	100%	90%	100%	100%	90%	90%
Subject9	100%	100%	100%	90%	90%	100%	100%	100%	90%	100%
subject10	90%	100%	90%	100%	100%	100%	90%	90%	100%	100%
Overall Average	96.7%									

contains samples of the 5's taken from user speech and converted to a spectrogram; the false class contain other numbers than 5, that are pronounced by the user.

5.2 Training, validation and results

With a learning rate of 0.001, the system is trained using stochastic gradient descent with momentum (SGDM). Table 1 shows the number of epochs and iterations.

In total of 4000 tests, including ten subjects with ten digits, the best training and validation accuracies are achieved. Each subject's sound sample from 0 to 9 is tested 4 times. At the beginning of every experiment, the overall data is shuffled and then training and validation split is done. This process is repeated 4 times and then the average rates are recorded in order to validate the results.

Table 2 displays the test accuracy for each individual vs. digit. The figures here are the average of four tests for each digit. The average test accuracy for each digit is provided in the same table. The system's validation accuracy is 96.7% on average.

Table 3 displays the validation accuracy for each subject vs. digit. The figures here are the average of four tests for each digit. The system achieves the average validation accuracy of 95.5%.

Although few digits are not pronounced good, the system is even capable to classify them as well. The results can be considered as satisfactory as in a conventional

Table 3 Validation accuracy of each subject/digit

Subject/Digit	0	1	2	3	4	5	6	7	8	9
subject1	100%	100%	90%	90%	90%	90%	100%	90%	100%	90%
Subject2	90%	100%	100%	100%	100%	90%	90%	100%	90%	100%
Subject3	100%	90%	90%	90%	100%	90%	100%	90%	90%	100%
Subject4	90%	90%	100%	90%	90%	100%	100%	100%	100%	90%
Subject5	100%	100%	90%	90%	100%	100%	100%	100%	100%	90%
Subject6	90%	90%	100%	100%	90%	90%	90%	100%	90%	90%
Subject7	100%	100%	90%	100%	100%	90%	100%	90%	100%	90%
Subject8	100%	100%	100%	100%	100%	100%	90%	100%	100%	100%
Subject9	90%	100%	90%	90%	90%	100%	100%	90%	100%	100%
subject10	100%	100%	100%	100%	90%	90%	90%	100%	100%	90%
Overall Average	95.5%									

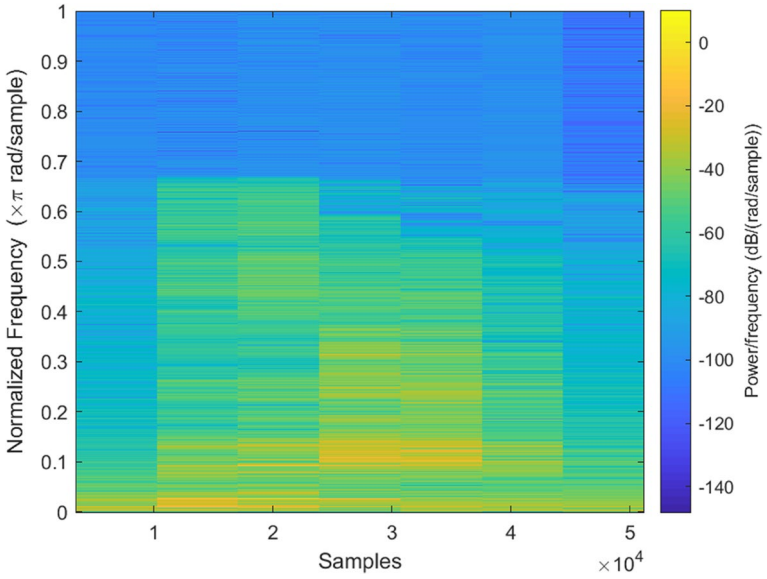


Fig. 10 A student recorded spectrogram sample of saying 5

voice recognition system the accuracies are even lower for some poorly pronounced digits (Fig. 10).

The pronunciation of the special education students is more difficult to understand compared to their agemates. Therefore, the system is trained for each student separately and it is subject dependent. Table 4 shows the comparison of the proposed study together with the other studies.

The proposed approach achieves comparable performances as shown in Table 4. The other state of the art studies listed in the table are dedicated to speech recognition, rather than verification. In (Nagajyothi & Siddaiah, 2018), the performances of

Table 4 Comparison table for the proposed approach

Authors	Name of Study	Method	Accuracy	Comparison with proposed method
Muhammad Musaev, Ilyos Khujayorov, Mannon Ochilov (Musaev et al., n.d.)	Image Approach to Speech Recognition on CNN	Speech recognition using spectogram images and deep convolution neural network of Uzbek spoken digits	100%	• Applied to Uzbek digits • Person independent CNN used • Speech recognition approach used • Rich dataset is used • It is applied on normal individuals
Tanaka et al. (2019)	A joint end-to-end and DNN-HMM hybrid automatic speech recognition system with transferring sharable knowledge	Frame-by-frame mannedr deep neural network hidden Markov model(DNN-HMM)	92%	• Stable processing based on a frame-by-frame manner • Japanese lecture corpus is used as dataset • AdaDelta algorithm to optimize the parameters • Applied for speech recognition
N. Dimmita, P. Siddaiah (2018)	Speech Recognition Using Convolutional Neural Networks	Convolutional Neural Network trained from scratch	Similarity Metrics are reported for the recognized words	• CNN is employed • Telugu language is used • Real recordings from airport • Person independent
Proposed Approach	Subject Dependent Speech Verification Approach for Assistive Special Education	Subject-dependent speech verification system is verifying the spech of special education students based on CNN and spectogram images	96.7%	• Applied to Turish digits • Subject-dependent CNN is used • Speech veriication approach is used • Limited dataset is used • It is applied to individuals who need special education

the standard Feed Forward Neural Network and the Convolutional Neural Network are compared for speech recognition. In (Tanaka et al., 2019), a hybrid Neural Network-Hidden Markov Model is employed. All the reported experimental results are seemed to be satisfactory but their problem domains are different. For instance, in Nagajyothi and Siddaiah (2018), the experimental data is collected from an airport. In (Ioffe & Szegedy, 2015), Japanese language is used. In this aspect, the originality of the proposed approach is that its domain is special education students.

A fair comparison of the proposed method is done with M.Musaev et al.'s study. They have also used a similar approach to recognize the Uzbek digits. Their training set is enriched, including the speech data of every individual subject, whereas, in our study, we only use students who need special education. Under this limitation, our system performs. On the other hand, our system is a verification system, whereas the compared system is a recognition system. Also, application to the Turkish language domain is another significant point to discuss. Under all the limitations and similarities, the proposed method achieves comparable and high rates.

6 Conclusion and future work

In this paper, a speech verification approach is proposed that is a person-dependent approach, in order to verify the digits from the speech of the special education students. Considering the marginality of the speech of individuals with special needs, the proposed approach fills in this gap and it is a positive addition to the literature in this aspect. The approach uses s a CNN. The CNN used is trained from scratch. The dataset used is another originality of the paper, which is accomplished by collecting the pronounced digits of the students under the supervision of special education experts. The overall approach verifies speech in a satisfactory level. The system is capable to achieve 96.7% average accuracy. We believe that, with a possible mobile application integration, the system will be able to support the teaching of special education students in the future.

6.1 Theoritical contribution

The proposed approach is contributing to the field in the way that it is employing deep learning models with spectrogram inputs of audio signals that are obtained from special education students. Furthermore, the approach contributes to artificial intelligence applications related to special education. Another aspect is the person dependency of the approach where each student is treated as separate sample and a dedicated deep learning model is trained for every student by using their own audio inputs. Considering all these aspects, the approach is a positive addition to the literature about employing artificial intelligence algorithms to special education domain.

6.2 Practical implications

The proposed approach will be useful in mobile or web based software applications in order to assist the teacher. As the system is able to automatically recognize the voices of the special education students and their digit pronunciations, it will be an assistive technology to be used by the teachers. In practice, the special education students need one to one sessions. In this sense, using an assistive application will enhance the teacher's performance. On the other hand, another practical implication is that the proposed approach can be a part of a possible mobile or web based application that will be directly used by the students under the supervision. Automatic audio verification tasks of such an application can be performed by using the proposed approach.

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Data Availability All data and materials used can be accessed through the authors of the paper. All data generated or analyzed during this study are included in this published article.

Declarations

Ethical approval This study is approved by the ethical committee of the institution where the study is taking place.

Competing interests There are no relevant financial or non-financial competing interests to report.

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